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# Ex - 6

## Implement rule based, pattern-based and machine learning algorithms for information extraction on clinical text and generate structured datasets.

## Aim : To implement **three different methods**(rule-based, pattern-based, and machine learning) for extracting structured data (Patient ID, Age, Diagnosis) from clinical text and generate structured datasets.

## Objective :

- **Rule-Based Mastery:** To understand and implement a simple approach using regular expressions for direct data extraction.
- **Pattern Flexibility:** To implement a pattern-based approach using keyword matching for flexible extraction from clinical text.
- **ML Integration:** To apply a machine learning approach (**Naive Bayes classifier**) to predict diagnoses from tokenized text features.
- **Comparative Analysis:** To compare the strengths, limitations, and outputs of all three approaches on the same clinical dataset.

## Procedure :

- **Step 1: Import Libraries** – Bring in the necessary Python libraries (`re` for regex, `CountVectorizer`, and `MultinomialNB` from `scikit-learn`).
- **Step 2: Data Preparation** – Create a sample list of clinical text records containing Patient ID, Age, and Diagnosis.
- **Step 3: Rule-Based Extraction** – Define the `rule_based_extraction` function using `re.findall()` with a precise regex pattern.
- **Step 4: Pattern-Based Extraction** – Define the `pattern_based_extraction` function using a list of keywords and `re.search()` for each field.
- **Step 5: Machine Learning Approach** – Define the `machine_learning_extraction` function:
  - Convert text to features using `CountVectorizer`.
  - Extract labels and train a `MultinomialNB` classifier.
  - Generate structured predictions.
- **Step 6: Execution** – Call each extraction function on the clinical data and store the results.
- **Step 7: Visualization** – Print the results of all three methods side-by-side for comparison.
- **Step 8: Evaluation** – Observe and discuss the differences in output format and performance.

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### Step 1: Import Libraries

We need regular expressions for rule-based and pattern-based methods, and scikit-learn tools for the machine learning approach.

```python
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB
import re
```

[1] ✓ 0.0s Python

## Step 2: Data Preparation

Create sample clinical notes in a consistent format.
This simulates the kind of semi-structured text we often see in medical records.

```python
clinical_data = [
    "Patient ID: 001, Age: 45, Diagnosis: Diabetes",
    "Patient ID: 002, Age: 60, Diagnosis: Hypertension",
    "Patient ID: 003, Age: 29, Diagnosis: Diabetes",
    "Patient ID: 004, Age: 40, Diagnosis: BloodPressure",
    "Patient ID: 005, Age: 52, Diagnosis: Depression"
]
```

[2] ✓ 0.0s Python

## Step 3: Rule-Based Extraction

Uses one strict regular expression to match the **entire** expected pattern.
→ Very fast, but fails if format changes even slightly.

```python
# def rule_based_extraction(data) :
#     extracted_results = []

#     pattern = 'Patient ID : (\d+), Age: (\d+), Diagnosis: (\w+)'

#     for record in data :
#         match = re.findall(pattern, record)
#         extarcted_results.append(match)

#     return extracted_results
```

[3] ✓ 0.0s Python

```python
import re

def rule_based_extraction(data):
    extracted_results = []

    # 1. Loop through each clinical record one by one
    for record in data:
        # 2. Define a pattern that looks for specific 'labels' in the text
        # (\d+) means 'find numbers', (\w+) means 'find words'
        pattern = r'Patient ID: (\d+), Age: (\d+), Diagnosis: (\w+)'

        # 3. Find all matches in the current record
        match = re.findall(pattern, record)

        # 4. Save the match to our list
        extracted_results.append(match)

    return extracted_results
```

[4] ✓ 0.0s Python

## Step 4: Pattern-Based Extraction

More flexible than strict rigid regular expression (regex) — looks for each field independently. Returns `None` if a field is missing → better at handling variation.

```python
def pattern_based_extraction(data) :
    patterns = ['Patient ID','Age','Diagnosis']
    extracted_data = []

    for record in data :
        record_data = []
        for pattern in patterns :
            match = re.search(f'{pattern}:(\w+)', record)
            record_data.append(match.group(1) if match else None)

        extracted_data.append(tuple(record_data))
    return extracted_data
```

[6] ✓ 0.0s Python

## Step 5: Machine Learning Approach (Naive Bayes)

- Converts text to word count features
- Uses the diagnosis as label
- Trains and predicts on the **same** small data (for demonstration only)
  → In real projects we use train/test split + more data

```python
def machine_learning_extraction(data):
    # Feature extraction
    vectorizer = CountVectorizer()
    X = vectorizer.fit_transform(data)

    # True labels (diagnosis)
    y = [record.split(", ")[2].split(": ")[1] for record in data]

    # Train Naive Bayes
    clf = MultinomialNB().fit(X, y)

    # Predict (on training data → should match exactly here)
    predictions = clf.predict(X)

    # Extract patient IDs for nice output
    patient_ids = [record.split(", ")[0].split(": ")[1] for record in data]

    return list(zip(patient_ids, y, predictions))
```

[7] ✓ 0.0s Python

## Step 6: Execution

Run all three methods and store results.

```python
rule_based_result = rule_based_extraction(clinical_data)
pattern_based_result = pattern_based_extraction(clinical_data)
ml_result = machine_learning_extraction(clinical_data)
```

[9] ✓ 0.0s Python

## Step 7: Visualization / Output

Print results from all three approaches for easy comparison.

```python
print("=== Rule-based Extraction ===")
print(rule_based_result)
print()

print("=== Pattern-based Extraction ===")
print(pattern_based_result)
print()

print("=== Machine Learning (Naive Bayes) ===")
print(ml_result)
```

[10] ✓ 0.0s Python

```
=== Rule-based Extraction ===
[[('001', '45', 'Diabetes')], [('002', '60', 'Hypertension')], [('003', '29', 'Diabetes')],

=== Pattern-based Extraction ===
[(None, None, None), (None, None, None), (None, None, None), (None, None, None), (None, Nor

=== Machine Learning (Naive Bayes) ===
[('001', 'Diabetes', 'Diabetes'), ('002', 'Hypertension', 'Hypertension'), ('003', 'Diabete
```

## Step 8: Quick Evaluation & Observations

- **Rule-based**: strictest, fastest, clean tuples/lists, fails on format changes
- **Pattern-based**: more tolerant to small variations, returns `None` when missing
- **Machine Learning**: most flexible in theory (learns patterns), but here overfits perfectly because we train & predict on same tiny data
  → In real life: ML needs much more data + train/test split + proper evaluation metrics

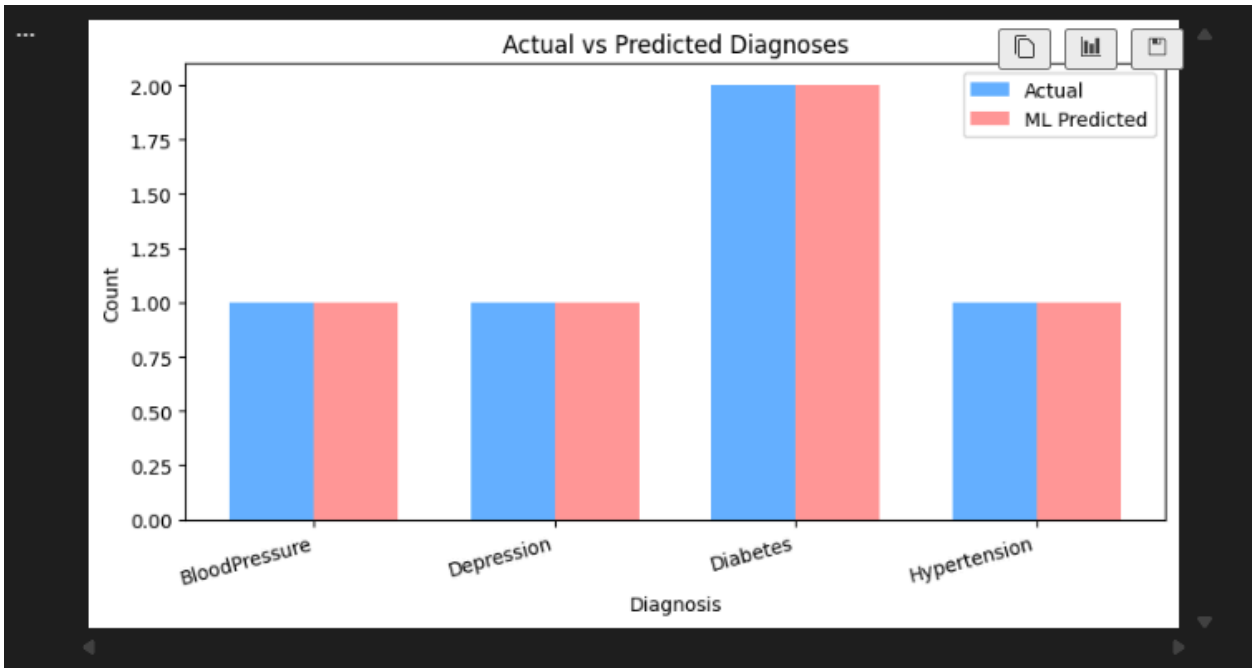
Which method would you choose for real hospital discharge summaries? Why?

EXTRA 👍

# Diagnosis Distribution – Actual vs ML Prediction

```python
from collections import Counter

actual_diags = [item[1] for item in ml_result]
pred_diags   = [item[2] for item in ml_result]

actual_count = Counter(actual_diags)
pred_count   = Counter(pred_diags)

all_diagnoses = sorted(set(actual_diags + pred_diags))

x = np.arange(len(all_diagnoses))
width = 0.35

fig, ax = plt.subplots(figsize=(8, 4.5))
ax.bar(x - width/2, [actual_count.get(d, 0) for d in all_diagnoses], width, label='Actu
ax.bar(x + width/2, [pred_count.get(d, 0) for d in all_diagnoses], width, label='ML Pre

ax.set_xlabel('Diagnosis')
ax.set_ylabel('Count')
ax.set_title('Actual vs Predicted Diagnoses')
ax.set_xticks(x)
ax.set_xticklabels(all_diagnoses, rotation=15, ha='right')
ax.legend()

plt.tight_layout()
plt.show()
```

[14] ✓ 0.1s Python


Actual vs Predicted Diagnoses
Actual
ML Predicted
Count
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
BloodPressure
Depression
Diabetes
Hypertension
Diagnosis